

QUIZ 02, VERSION X – Solutions

STAT 1110: Section X – Fall 2026 with Dr. Ethan P. Marzban

NAME: _____

First and Last

USER ID: _____

This is what comes before the @ in your Cal Poly Email Address

Instructions:

- You will have **20 minutes** to complete this **15-question quiz**.
 - The use of calculators is permitted, but the use of any other aids (including, but not limited to: laptops, cell phones, and note sheets) is prohibited.
 - **Please do not remove any pages from this exam booklet.**
 - Mark all of your answers on the provided bubble sheet; **nothing written in this exam booklet will be graded.**
 - Please make sure to write your **FULL NAME** name in the appropriate space on your bubble sheet.
 - In the “ID” space of your bubble sheet, please write your User ID (NOT your Student ID).
 - In the “Section” space of your bubble sheet, please write **X**.
 - Please also make sure to **indicate your Quiz Version** on your bubble sheet as well. You have **Version X**
 - Failure to correctly fill in the information on your bubble sheet may result in your quiz being unable to be graded.
 - Kindly note that partial credit is not available, as all questions are multiple choice.
 - Unless otherwise specified, each question is worth **one point**.
 - Good Luck!
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Honor Statement

In signing my name below, I certify that all work appearing on this quiz is entirely my own and not copied from any external source. I also certify that I have not received any unauthorized aid (including generative AI tools), and that I have acted in accordance with the Cal Poly standards of academic integrity. I acknowledge that failure to abide by these terms will be treated as academic misconduct.

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Use the following setup for Questions 1 through 9:

Rain wants to determine the proportion of Cal Poly students that own an electric scooter. They consult the registrar's office and obtain a list of the email addresses of all currently-enrolled Cal Poly students, and use this to assign each Cal Poly student a unique number. They then use a random number generator to randomly generate 20 numbers, and look up the emails of the 20 Cal Poly students corresponding to these 20 numbers and send them an email survey asking whether or not they own an electric scooter. All 20 students respond; 15 of them report owning an electric scooter.

1. Describe an observational unit in the context of this dataset.

(A) **One person**

(B) One electric scooter

(C) Whether or not one person owns an electric scooter

(D) Cal Poly

Solution: Rain collects information (specifically, about electric scooter ownership) from individual people (Cal Poly students) - hence, an observational unit is **one person**.

2. Describe the population of interest.

(A) The 20 people who responded to Rain's survey

(B) **The set of all Cal Poly students**

(C) The set of all Cal Poly students who own an electric scooter

(D) Rain

Solution: The population is, by definition, the set of all observational units of interest. Rain wants to determine something about the set of all Cal Poly students (specifically, the proportion of these that own an electric scooter) - therefore, the population is **the set of all Cal Poly students**.

3. What is the parameter of interest?

(A) A population mean

(B) A population median

(C) A population standard deviation

(D) **A population proportion**

(E) Something else

Solution: Rain wants to determine the *proportion* of the population (Cal Poly students) that own an electric scooter - therefore, the parameter of interest is a population **proportion**.

4. The quantity 15/20 is important to this problem. Is it a statistic value or a parameter value?

- (A) **Statistic** (B) Parameter

Solution: A statistic is a numerical descriptor of some aspect of a sample. The value of 15/20 is the *sample proportion of students in Rain's sample* that own an electric bike. Hence, it is a **statistic** value.

5. Which of the following best describes the sampling frame in the context of this study?

- (A) The set of people that responded to Rain's survey.
(B) **The set of all Cal Poly students**
(C) The set of all Cal Poly students who own an electric scooter
(D) Something else

Solution: We are told that Rain obtains a list of all registered Cal Poly students; as such, any Cal Poly student could have theoretically been included in Rain's sample. Therefore, the sampling frame is the same as the population in this case - the **set of all Cal Poly students**.

6. Suppose we determined that the sampling frame is the same as the population in this study (which is not necessarily the correct answer to Question (5) above!). Did Rain take a simple random sample?

- (A) **Yes; because they used a random number generator, any sample of size 20 was equally likely to have been taken from the sampling frame (which is, by assumption, equal to the population), and therefore they have taken a simple random sample.**
(B) Yes; because they are sending their survey through email, which is a secure form of communication, we can be assured that the results are legitimate and therefore came from a simple random sample.
(C) There isn't enough information to determine; because Rain used a random number generator, there is some uncertainty as to whether or not they took a simple random sample.

Solution: If the sampling frame and population are equal and if we use a random number generator, we are (by the definition of simple random sampling we're using at this point in the course) guaranteed to have taken a simple random sample. Since Rain did exactly that, they took a simple random sample.

7. Suppose that, instead of all 20 people replying to Rain's survey, only 17 responded. The fact that 3 people did not respond is an example of what type of bias, discussed in Lecture Activity 3?

- (A) Systematic bias
(B) Intrinsic bias
(C) Probabilistic bias
(D) **Non-response bias**

Solution: Non-response bias arises when certain members of the sampling frame are attempted to be included in a sample, but choose not to respond (or have their data dropped/lost).

8. Suppose that, unbeknownst to them, the registrar actually only gave Rain a list of emails of Freshmen and Sophomore. Further suppose that Freshmen and Sophomore are less likely to own electric scooters than Juniors and Sophomores, because they tend to live closer to campus. What type (if any) of bias might Rain's sampling method suffer from?

- (A) **Biased with a tendency to underestimate**
- (B) Biased with a tendency to overestimate
- (C) Rain's sampling method would be unbiased.

Solution: Since Rain's survey only reached people (Freshmen and Sophomore) who are less likely to own electric scooters, their results will likely undersell the true proportion of Cal Poly students that own electric scooters. That is, their sampling method will likely **underestimate** the true value of the parameter.

9. Rain believes their sampling method may be biased (which isn't to say this is the correct answer to any of the above questions!). As such, they are considering taking a much larger sample to compensate. Will this correct for the bias in the sampling method?

- (A) Yes; increasing the sample size will typically reduce the effects of bias.
- (B) Yes, but it depends as Rain would need to use a very large sample size to overcome the bias
- (C) **No; bias is something that's tied to a sampling method and cannot typically be eliminated by simply increasing the sample size.**

Solution: Since Rain's survey only reached people (Freshmen and Sophomore) who are less likely to own electric scooters, their results will likely undersell the true proportion of Cal Poly students that own electric scooters. That is, their sampling method will likely **underestimate** the true value of the parameter.

Use the following setup for Questions 10 through 15:

At a local county fair, a magician claims to be able to correctly guess a person's birthday just by looking at them. Indeed, out of 15 people, the magician correctly guessed the birthdays of 10 of them.

10. How can you describe this study?

- (A) Sample from a Population
- (B) **Observations from a Random Process**

Solution: There are a couple of ways to think about this. Firstly, there’s no “population” to speak of in this study (at least, not a finite one). Furthermore, each observational unit is “the result of the magician guessing a person’s birthday,” which is clearly one trial of a random process. So, either way we view it, our data takes the form of **observations from a random process**.

11. What is the statistic you can calculate from the sample?

- (A) The sample mean**

Solution: We can calculate the *proportion* of people whose birthdays the magician correctly guessed.

12. What is the value of the statistic?

- (A) 0 (B) 0.6667 (C) 5 (D) 15

Solution: The proportion of people whose birthdays the magician correctly guessed is $10/15 = 0.6667$.

13. Which of the following best describes the parameter of interest?

- (A) One person
(B) One person's birthday
(C) The long-run proportion of times the magician correctly guesses a person's birthday
(D) The long-run proportion of times someone has the same birthday as the magician

Solution: When our data represents observations of a random process, the parameter of interest is a long-run numerical summary of the underlying process. In the context of this study, this is the long-run proportion of times the magician correctly guesses a person’s birthday since the random process is “the outcome of the magician guessing a person’s birthday.”

14. Suppose it is known (through magic!) that the true probability the magician correctly identifies a person's birthday just by looking at them is 0.75. Which of the following provides the correct interpretation of the value 0.75? Select only ONE option.

- (A) Out of every 100 people, the magician will correctly guess the birthdays of exactly 75 of them just by looking at them.
- (B) In the long run, we'd expect the magician to correctly identify a person's birthday just by looking at them around three-fourths of the time.**
- (C) At next year's county fair, the magician will be able to identify around 75% of attendees' birthdays just by looking at them.

Solution: Answer choice (A) is too strong; remember that probabilities are not guarantees! Answer choice (C) is missing the crucial point that probabilities are *long-run* proportions; simply talking about “next year’s fair” doesn’t convey the long-run nature of the probability. Answer choice (B) is the best description of how to interpret the stated probability value.

- 15.** The magician claims that the true probability of him correctly guessing a person’s birthday is 1.45 (i.e. 145%). Is there anything wrong with this assertion? Select only ONE option.
- (A) No, there’s nothing wrong with this assertion.
 - (B) Yes, there’s something wrong with this assertion. The evidence (that he guessed only 10 out of the 15 people’s birthdays correctly) suggests that the probability should be lower.
 - (C) Yes, there is something wrong with this assertion, because probabilities cannot be greater than 1.0 (i.e. 100%).**

Solution: Probabilities must be between 0 (i.e. 0%) and 1 (i.e. 100%).
